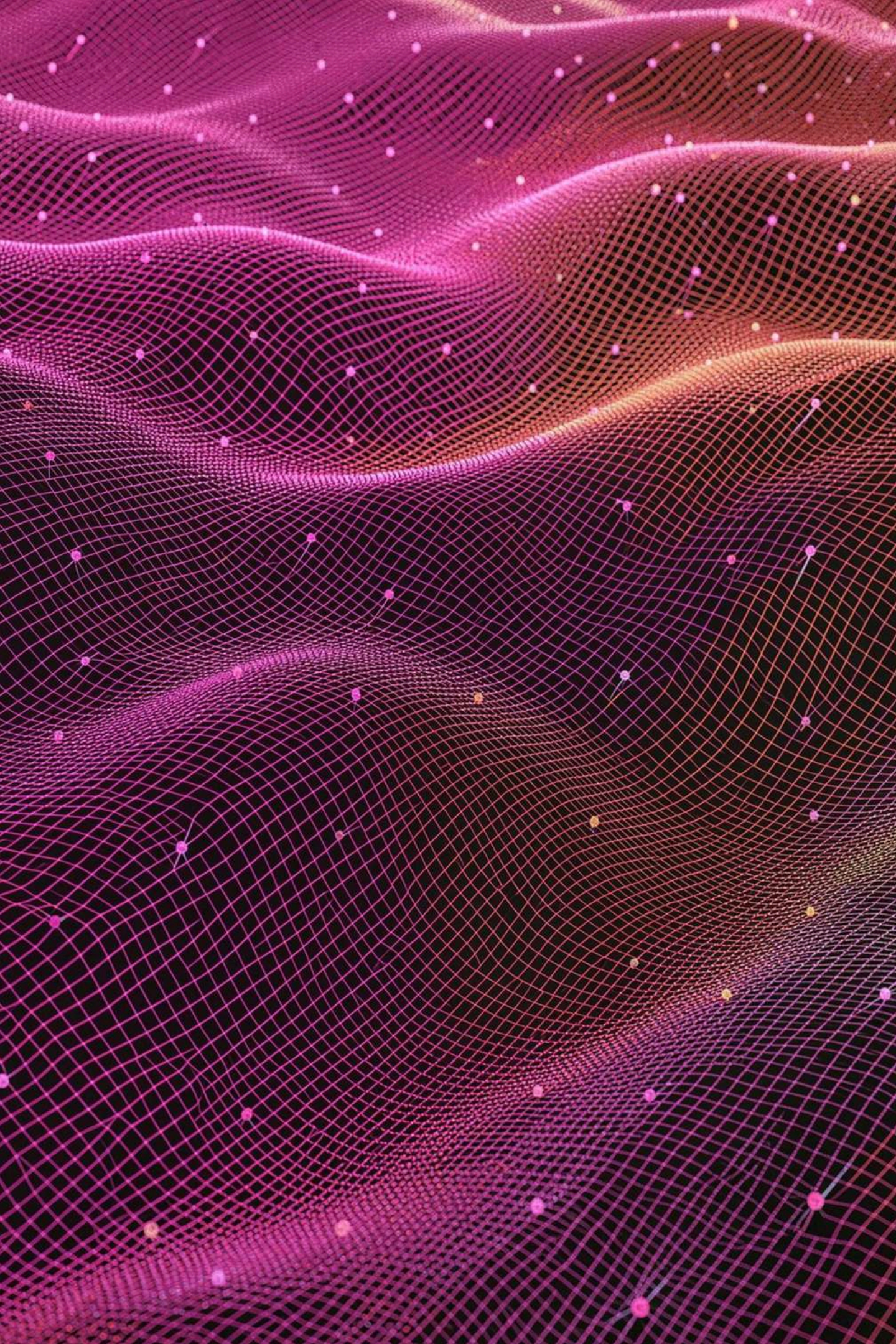
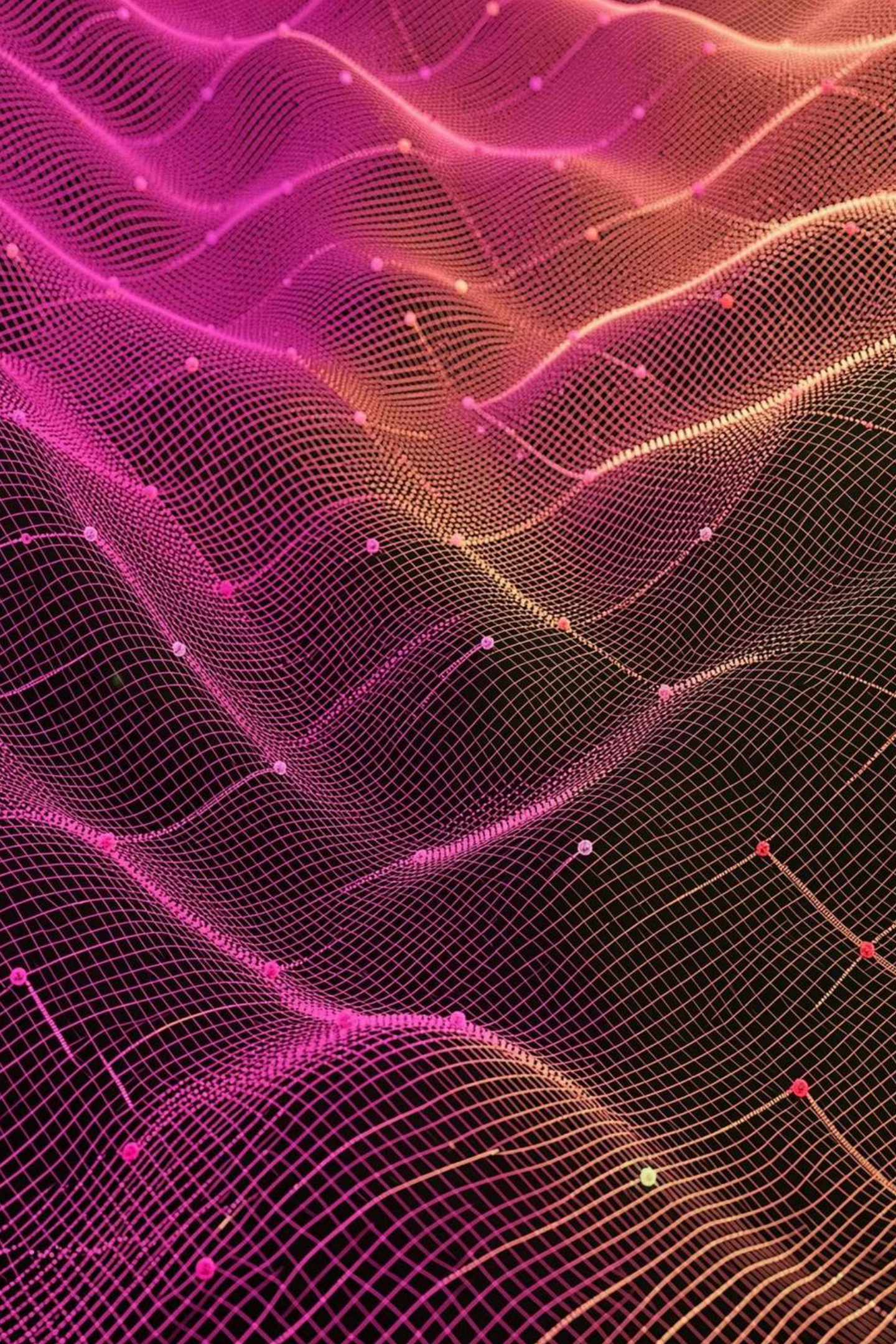




AI-Powered Fake News Detection Using Text Classification

Applying machine learning and natural language processing to combat misinformation at scale.



The Problem: Misinformation at Scale

The internet and social media have democratized information access, but also accelerated the spread of misinformation. With millions of articles published daily, manual fact-checking is no longer feasible.

Public Impact

Fake news misleads the public and influences opinions on critical issues — from elections to public health decisions.

Need for Automation

An automated classification system is essential for real-time verification at the scale modern media demands.

Project Objectives

This project applies machine learning and natural language processing to automatically classify news articles as fake or real, enabling fast and reliable verification.

1

Automated Classification

Develop a machine learning system to classify news articles without manual review.

2

Text Feature Extraction

Preprocess text and convert it into numerical features using TF-IDF vectorization.

3

Model Comparison

Train and evaluate multiple ML algorithms to identify the best-performing approach.

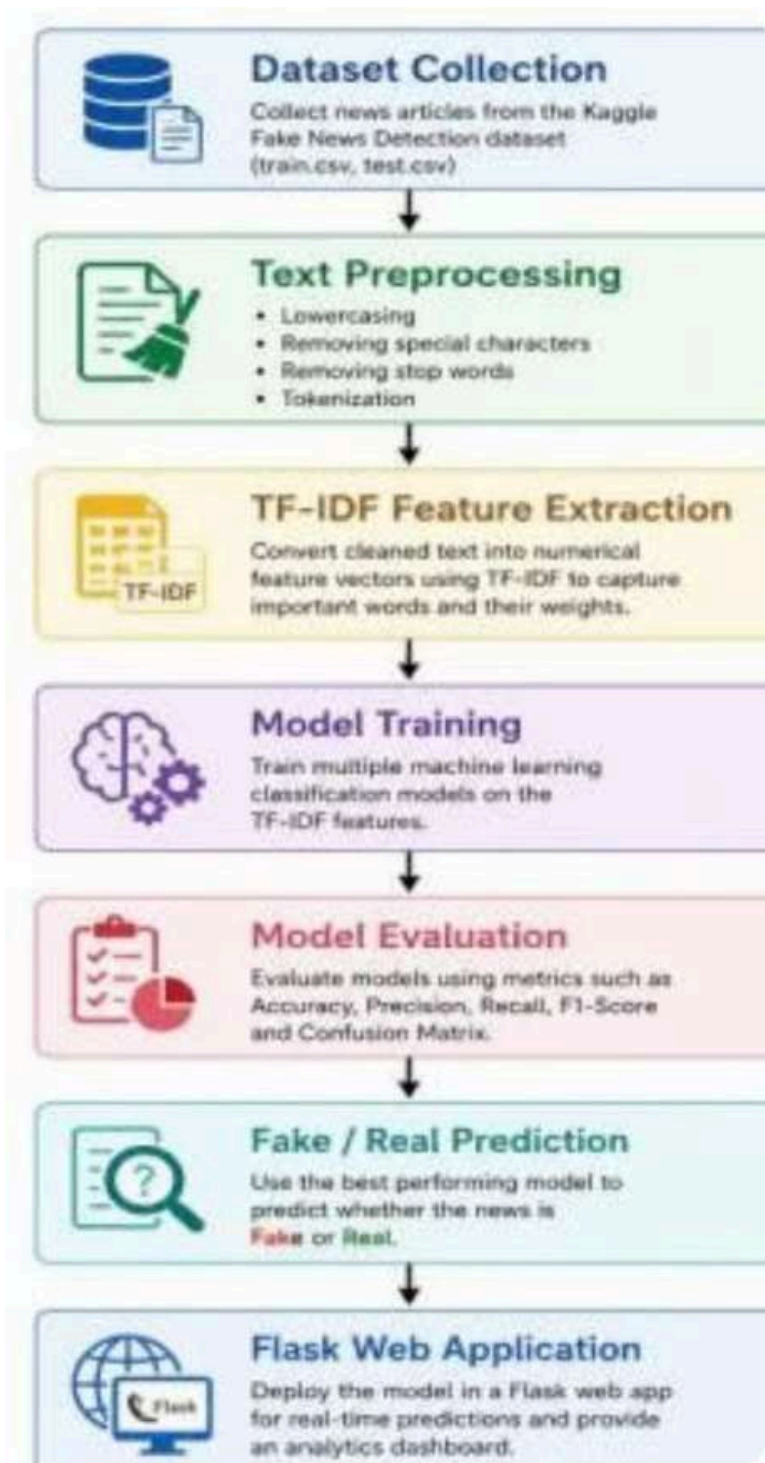
4

Web Deployment

Build a Flask web app for accessible real-time predictions with a user-friendly interface.

System Methodology: 7-Step Pipeline

The system follows a structured machine learning pipeline. Raw news articles are collected, cleaned, converted into numerical features, used to train multiple models, and the best model is deployed for real-time predictions.



01	Dataset
	Kaggle Fake News Detection Dataset (train.csv, test.csv)
02	Preprocessing
	Clean text, remove noise, tokenize
03	Feature Extraction
	TF-IDF converts text to numerical vectors
04	Model Training
	Compare KNN, Logistic Regression, Random Forest, Neural Network
05	Deployment
	Flask web app for real-time predictions

Data Processing & Feature Extraction

Raw text contains noise that reduces model accuracy. Preprocessing standardizes the data by removing irrelevant characters and words. TF-IDF (Term Frequency-Inverse Document Frequency) then converts cleaned text into numerical vectors, assigning higher weights to words that are important for distinguishing fake from real news.



Preprocessing

Lowercase, remove special characters, eliminate stop words, tokenize

TF-IDF Weighting

Captures word importance within articles and across the dataset

Output

High-quality numerical feature vectors ready for model training

Machine Learning Models & Evaluation

Classification models learn patterns from training data to predict whether unseen articles are fake or real. Multiple algorithms are trained and compared using key evaluation metrics.



Models Trained

- K-Nearest Neighbors (KNN)
- Logistic Regression
- Random Forest
- Neural Network

Evaluation Metrics

- **Accuracy** — Overall correctness
- **Precision** — How many predicted fakes are actually fake
- **Recall** — How many actual fakes are caught
- **F1-Score** — Balance between precision and recall
- **Confusion Matrix** — Visual error breakdown



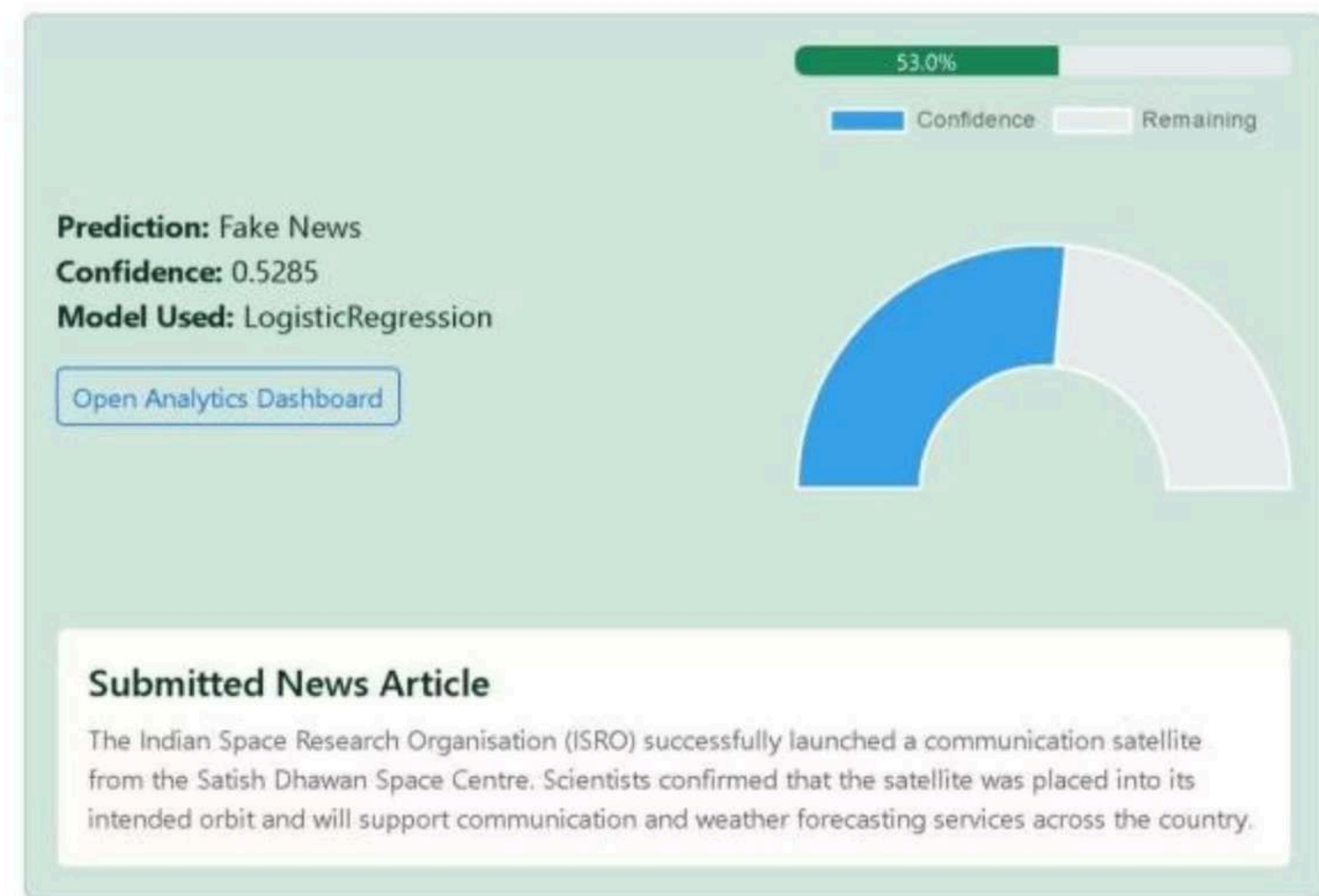
The best-performing model is selected and deployed in the Flask application.

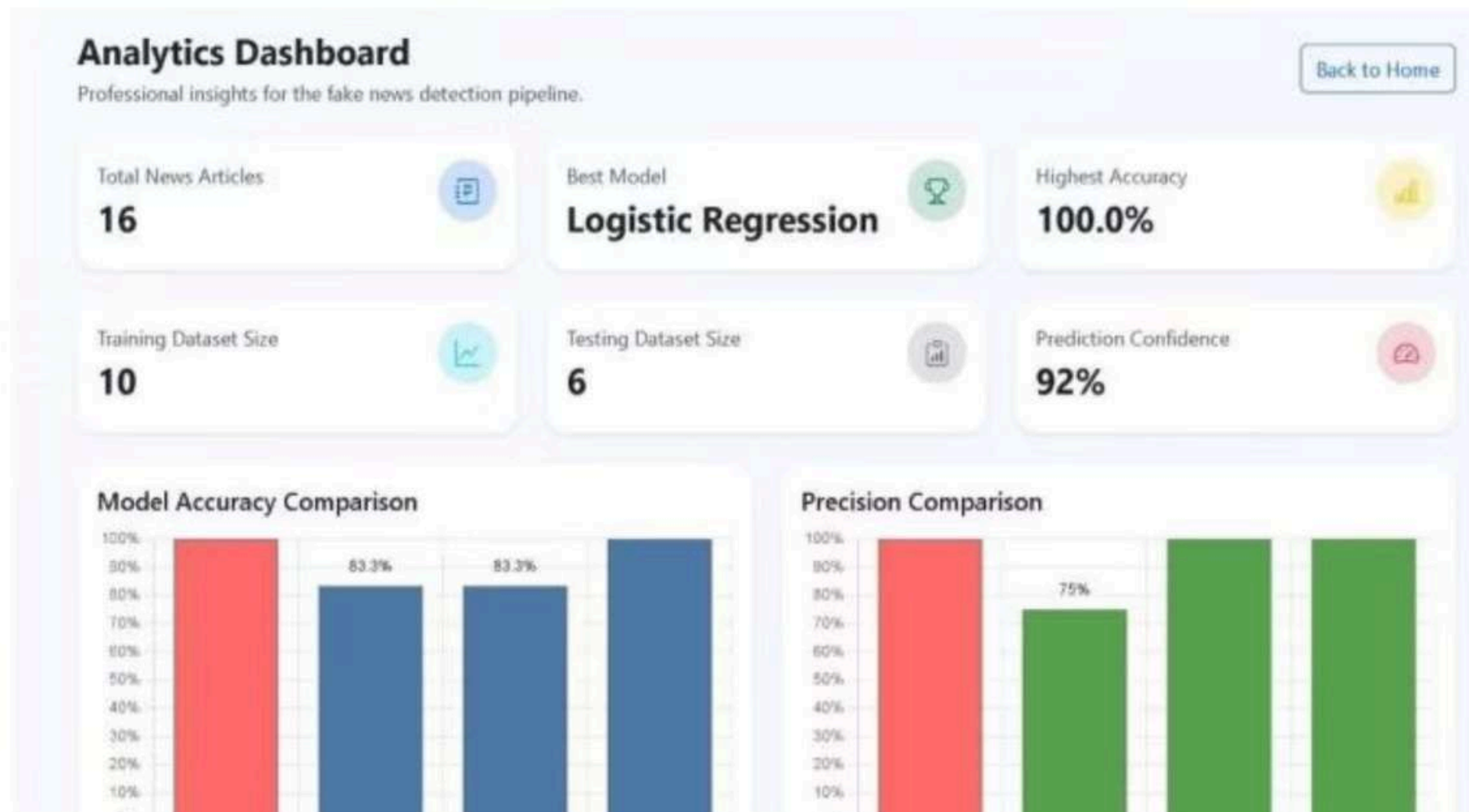
Web Application & Real-Time Predictions

The trained model is deployed in a Flask web application, making fake news detection accessible to end users. Users submit news articles through a simple interface and receive instant predictions with confidence scores.

How It Works

- **User Input**
News article text submitted via web form
- **Real-Time Output**
Fake/Real classification with confidence percentage
- **Analytics Dashboard**
Model performance metrics, confusion matrix, ROC curves, and dataset visualizations





Conclusion

This project demonstrates the practical application of machine learning and NLP to address the critical challenge of misinformation — providing fast, reliable, and scalable verification tools that process thousands of articles in real-time.

✓ High Accuracy

Successfully developed an automated classification system with strong performance across all metrics.

📊 Interactive Analytics

Dashboard enables continuous performance monitoring and actionable insights.

🚀 Future Scope

Modular design supports transformer-based models, cloud deployment, and multilingual support.

Thank You

